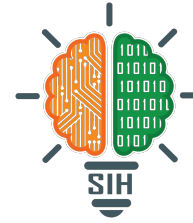


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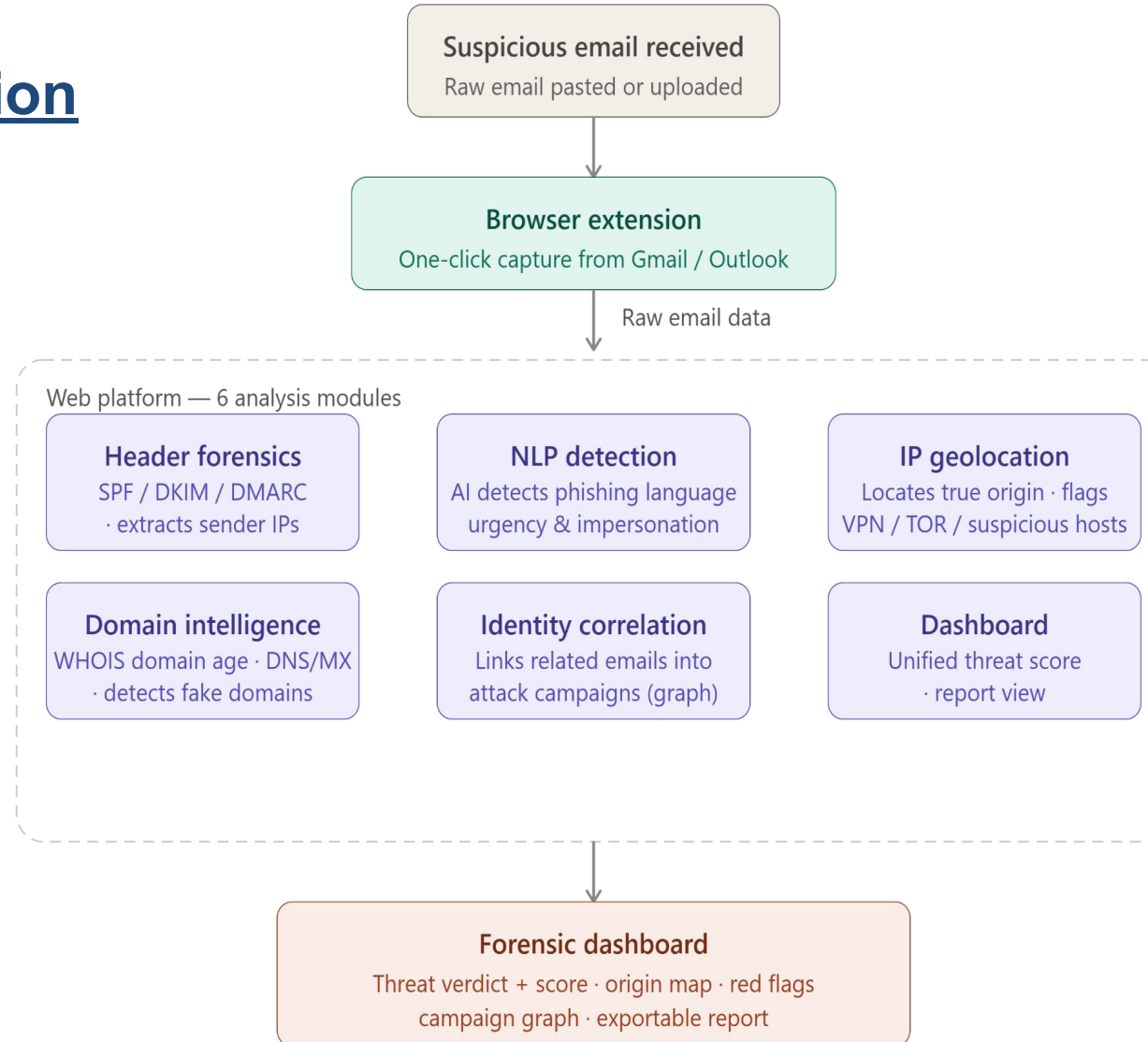
- **Problem Statement ID – SIH26106**
- **Problem Statement Title-** AI-Powered Email Threat Detection, GeoLocation and Forensic Intelligence Platform
- **Theme-** Blockchain & Cybersecurity
- **PS Category-** Software
- **Team ID-**
- **Team Name -** FraudBusters



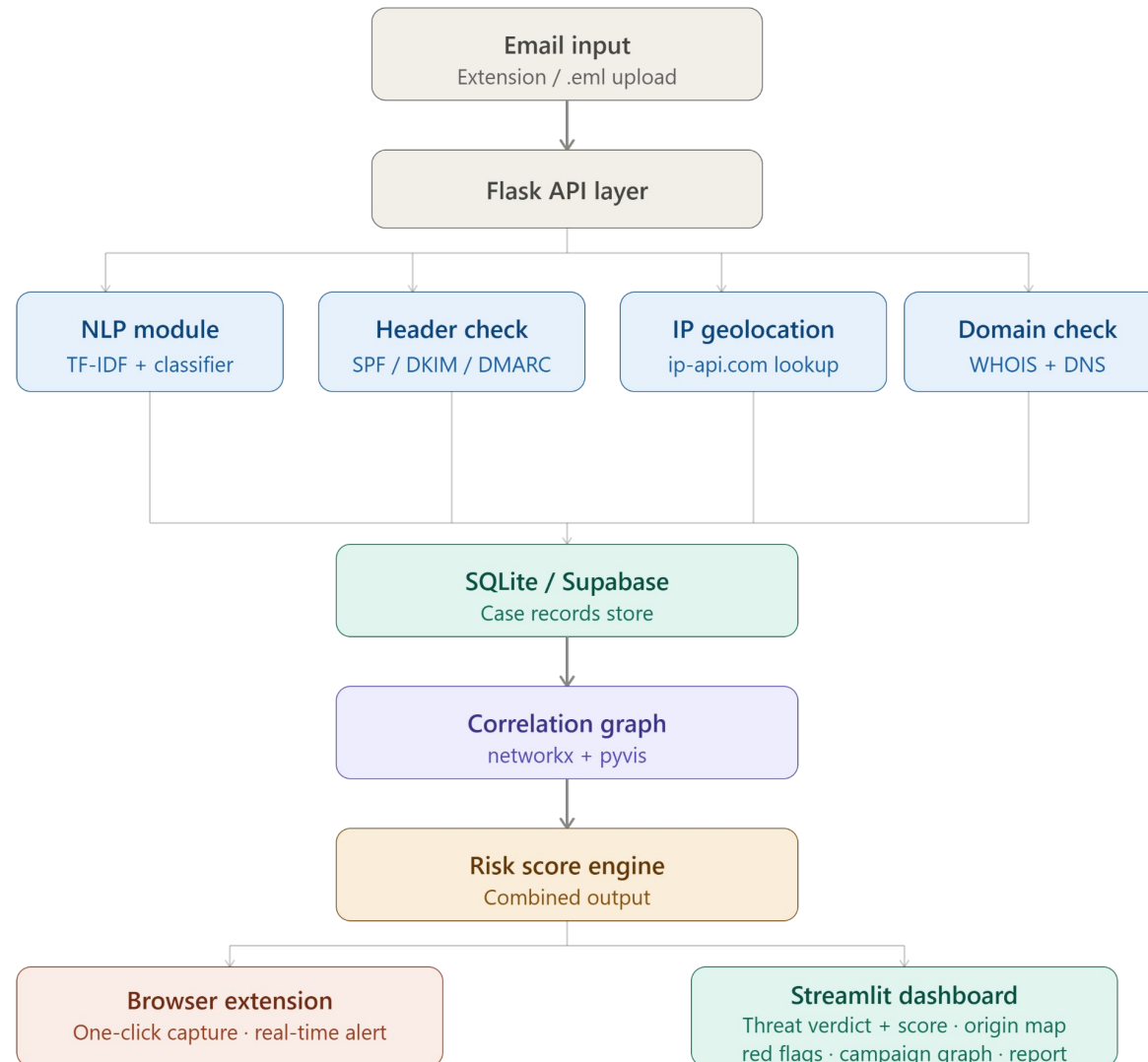
PhishIt:

Don't take the bait, Trace the Threat

Proposed Solution



TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

FEASIBILITY

- Built entirely with Python, AI/ML and free open-source tools
- Existing APIs supply IP geolocation and threat intelligence
- Trained on publicly available, labelled phishing email datasets
- Modular, web-based architecture — scalable and easy to extend
- Runs on standard hardware — no specialized equipment needed
- Achievable by a student team within the hackathon timeline

CHALLENGES

- Forged, missing or incomplete email headers
- VPN / Proxy / TOR can mask the attacker's true source
- IP geolocation gives approximate, not exact, location
- Free API rate limits and usage restrictions
- Handling large volumes of emails efficiently
- Keeping pace with constantly evolving phishing techniques

RISKS

- ML false positives (flagging safe emails as threats)
- ML false negatives (missing real phishing)
- Over-reliance on approximate geolocation data
- Privacy concerns when handling real email data
- Dependency on third-party APIs (downtime or changes)
- Attackers adapting to evade detection

STRATEGIES (Solutions to Challenges)

- Use advanced header forensics and anomaly detection to handle forged or incomplete headers
- Analyze multiple signals (headers, content, behavior) to reduce the impact of VPN / Proxy / TOR
- Use confidence scoring for geolocation and combine with other indicators for better accuracy
- Implement API caching, quota management and alternative free sources to overcome rate limits
- Optimize processing with queuing, batching and asynchronous tasks for high email volumes
- Continuously update threat intelligence and retrain models to detect new phishing techniques

IMPACT AND BENEFITS

- **Users:** Police/cybercrime cells, government, banks, citizens
- **Social:** Shields citizens from phishing-driven fraud; builds digital trust
- **Economic:** Prevents financial losses; saves investigation time & cost
- **Security:** Traces, attributes & links attack campaigns; boosts national cyber-defense
- **Environmental:** Paperless, resource-efficient forensics

1. **MITRE ATT&CK – Phishing (T1566)**

Legit and highly relevant to your project. It supports your research on phishing, impersonation, malicious links, spoofing and email-based attacks.

[MITRE ATT&CK – Phishing T1566](#)

2. **MITRE ATT&CK – Spearphishing Link (T1566.002)**

Very relevant for your malicious/hidden URL detection component. MITRE also discusses URL obfuscation and lookalike/homograph domains.

[MITRE ATT&CK – Spearphishing Link](#)

3. **MITRE ATT&CK – Phishing Detection Strategy (DET0070)**

This is **especially useful for your project** because it explicitly describes correlating email metadata, URLs, files and network activity for phishing detection.

[MITRE ATT&CK – Phishing Detection Strategy](#)

4. **NetworkX Documentation**

Legitimate reference for the graph-analysis component you're using to model relationships between emails, domains, IPs and URLs.

[Network Documentation](#)